A star is born

Like many space pictures, this image of the Eagle Nebula has been artificially coloured so it can be seen clearly.

Clusters of stars are constantly being born from clouds of gas and dust thousands of times the size of our solar system, in a process that can take millions of years.

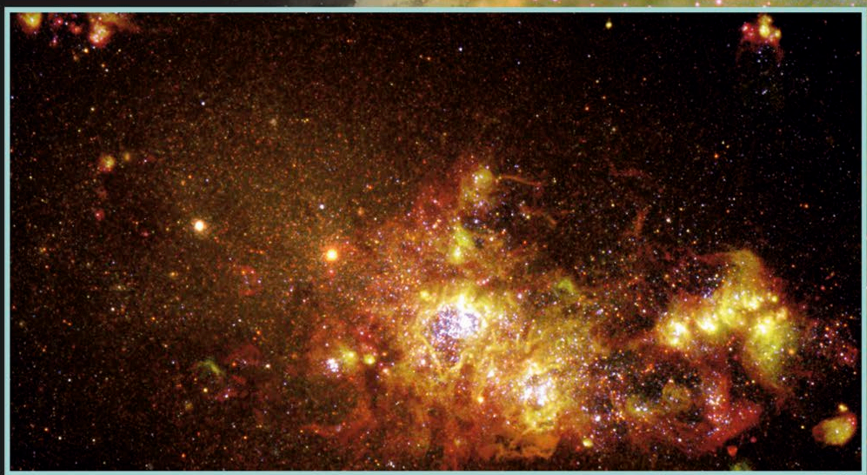
Born in a cloud

Between existing stars, there are patches of gas and dust. Gradually, these draw in more and more gas and dust to form huge clouds called nebulae.

Nebula

Hot colours

As the matter within gets more and more dense, the clouds start to shrink under their own gravity, and eventually break into clumps. This collapse builds up heat, which forms a dense, hot core (protostar) that fills the surrounding nebulae with light and colour. This spectacular effect (right) was captured by the *Spitzer* space telescope.



The process of star formation captured by the *Hubble* telescope.

We have fusion!

With enough matter, the process of heating continues. The core gets denser and hotter. Eventually nuclear fusion begins, releasing huge amounts of heat and light – a star is born.

Which star cluster is also called the Seven Sisters?